6

Necessary Conditions for Democratic Reform

This chapter presents the analysis of necessary conditions, and shows that strong claims about finding necessary conditions in PB research have been premature. They are likely to have been an artefact of small-N research strategies where deviant cases that reduce evidence for necessity are not as obvious as they become when comparing the full range of evidence across research studies. There is quite a lot of evidence to show meaningful citizen control occurs without autonomous civil society demand and organizing, bureaucratic support for programmes, and strong financial resources respectively. That is not to say that case researchers were not on the right track with their claims, only that these claims need some further disambiguation to better identify the scope or context in which a condition becomes necessary in combination with other conditions. This is a crucial finding because it suggests that there is no context that is universally and unambiguously unable to produce some good PB outcomes, and therefore there is no place where PB will always fail (at least in the short term). Nevertheless, there is strong evidence that political leaders' commitment to participation is almost always necessary for success.

A researcher's task in presenting research is to distil excessive complexity and explain social phenomena with appropriate levels of parsimony. To this end researchers often construct typologies, and make claims about necessary and sufficient conditions based on their observations. Brian Wampler applies a familiar controlled approach in his distinguished comparison of eight cases of PB in urban Brazil (2007a). This is a useful contribution for scholars who wish to explain the emergence of emancipatory participatory governance through democratic innovation. It has been commended not only for its attentive and detailed narrative of processes within cases, but also its comparative design that moved beyond

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best-case examples to include variations in outcome and explanatory conditions to explain causation more systematically. In common with other examples in the emerging field of research on participatory democratic innovations, comparative work up to now has relied on more traditional small-N comparative designs. Therefore, in this final part of the book I often introduce stages of the research process by presenting a version of what a QCA replication of that study might look like. I then present wider analysis that accumulates more cases across studies. I take this approach to help readers understand and evaluate what the method achieves and what trade-offs are involved in taking on more set-theoretic reasoning over a larger-N of cases.

Notwithstanding Flyvbjerg's assertion (2006) that admonitions on subjectivity essentially misunderstand the reflexive nature of case-based research, a replication controlling even for the researcher's case choices but testing the added value of a QCA approach can be useful. Developing on increasing calls for transparency of data sources for replicability from case researchers, I implore that a QCA approach can also recover transparency at the analytic moment of smaller-N research by outlining what is at stake in decisions to highlight particular causal condition(s). I show that I can, at the very least, provide greater clarity when selecting the best trade-off between parsimony and explanatory power in small-to-medium-N case comparisons. I therefore directly address some of the possibilities of QCA as a deductive tool that aids interpretation where the case researchers may have their eyes very close to the field or data. The main model I present throughout these chapters is the result of careful reflection on which cases have provided enough information to make valid judgements for comparison.

It is important that readers unfamiliar with the method can understand the evidence base for my claims. I begin here by taking the example of a more parsimonious claim from Wampler's work to evaluate and explain the analysis and some pitfalls. Moreover, if I can add value to some of the most extensive existing studies, it provides support for the value of wider comparisons using such an approach.

Note again, then, that often Wampler's arguments are set-theoretic: 'To produce a strong PB programme, it is necessary to have high levels of mayoral support, a civil society that can engage in both cooperation and contestation, and rules that delegate specific types of direct authority to citizens' (2007a: 35).[1] To test these claims, I apply

[1] Wampler follows this quote directly with a correlational claim – 'as Mayoral support drops, as CSOs are unable to engage in both forms of political behaviour,

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QCA's systematic approach, calibrating conditions based on the same qualitative descriptions used in the existing comparative study.

Table 6.1[2] is adapted from Wampler's concluding chapter (2007a: 258). It is typical of the kinds of parsimonious typological tables often presented in the findings of small-N comparative work. These tables are often used as methodological heuristics to make set-theoretic claims of necessity and sufficiency.

In Wampler's concluding chapter, he places particular emphasis on two explanatory conditions and suggests that mayoral support interacts with CSOs' willingness to use contentious politics to give four very different types of outcomes: institutionalized participatory democracy, informal and contested participatory democracy, co-opted participatory democracy, and emasculated participatory democracy. This type of table will be familiar to small-N case researchers. This is because the major advantage of small-N research is underlying familiarity with cases. This allows Wampler to verbally define each box in the table where we have an empirical example of a case. That is, he is labelling a set of cases and identifying members of the set. This is a useful process that adds plausibility and clarity to the argument; for example, it makes intuitive sense that co-optation in participatory programmes can be traced back to low use of contentious politics by CSOs, as in Table 6.1. Wampler argues from the outset that citizens need to be able to actively pursue the use of their rights (2007a: 2), and his evidence bears that out. The researcher here has helped the reader to comprehend a complex process. Prima facie, it would seem fair to conclude that the institutionalization of deep democracy can *only* occur where both these conditions are high, and that both are necessary conditions for this outcome. A simple correlation between each variable and the outcome deep democracy would be positive. From the point of view of categorical and ordinal understandings of the relationship among the conditions reflected in the data, the typology is both informative and unproblematic. But with QCA we have a further test we can use to assess the necessity claim. So what can set-theoretic comparison add?

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and as the rules fail to delegate authority, PB outcomes will weaken' (2007a: 35). These two claims are not necessarily incompatible; what is important is that they require a different test. For the most part, both the correlational and set-theoretic inferences Wampler makes stand up well to scrutiny.

[2] The adaptation includes a split of Porto Alegre into three cases reflecting three time periods, which Wampler suggests reflects better the type of PB within the case at different times.

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Table 6.1: Types and causes of outcomes

		CSOs' willingness to use contentious politics	
		High	Low
Mayoral support for delegation of authority to citizens	High	Porto Alegre 1997–2004, Ipatinga [institutionalized participatory democracy]	No case
	Medium	Recife, Belo Horizonte, Porto Alegre 1989–96, Porto Alegre 2004 onwards [informal and contested participatory democracy]	Santo Andre, São Paulo [co-opted participatory democracy]
	Low	No case	Blumenau, Rio Claro [emasculated participatory democracy]

Source: adapted from Wampler (2007a: 258)

In order to elucidate the importance of calibration decisions and how our concepts affect our causal claims, it is useful to look at how differences between calibrations of set memberships used in interpreting nominal or ordinal data can affect our inferences about necessity. In Table 6.2 I calibrate some fuzzy sets to represent different interpretations of Wampler's categorical data presented in Table 6.1. Table 6.2 includes three different possible conceptualizations or measures of the outcome in order to discuss the effect of more significant differences in interpretation by applying fuzzy calibration and logic in the discussion. The first column from the left distinguishes the cases. The second to fourth columns represent three fuzzy sets for the two influencing conditions (mayoral support and citizen contentious politics), and one possible outcome ('deepened democracy A'). High presence of a condition according to Wampler's description is coded as full membership in the set (fuzzy set membership score

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of 1) and low as full non-membership (set score of 0). Medium membership is coded as 0.51.[3]

One might observe that the use of the word 'medium' by Wampler does not necessarily indicate halfway 'in' the concept. Equally, the word 'low' is hardly commensurate with the idea of complete non-membership of a set. This serves to highlight that there is some theory and interpretation involved in the process, and the fuzzy calibration is adding a layer of meaning. Fuzzy sets should not be confused with ordinal variables – to undertake set-theoretic analysis an ordinal variable will need to be converted to a fuzzy set. Small changes in the membership value will not fundamentally affect the subset–superset relations in QCA analysis. As a robustness test marginal changes in value can be examined to confirm insubstantial effects on the outcome.

I add some alternative calibrations of the outcome in the fifth and sixth columns to show how these choices matter. My purpose in this example is not to be precise in measurement, but to show the advantages and requirements of set-theory for understanding claims of necessity and sufficiency. For the time being, we can say that the original researcher has made a qualitative distinction of an ordinal kind (at least for the two influencing conditions), which has contributed to the calibration of a fuzzy set. For the outcome condition, Wampler's table really suggests four nominal categories, which, in some cases, I have then interpreted as ordered degrees of membership in at least three of the calibrations presented. I now show how this matters for tests of necessity.

In all cases in Table 6.2 the outcome condition representing deepened democracy codes cases having full membership (1) for those cases of institutionalized participatory democracy, and 0 for emasculated participatory democracy. The outcomes for the other cases vary. For 'deepened democracy A', I ascribe an outcome set membership of 0.51 for both categories of informal and contested participatory democracy

[3] Readers familiar with fsQCA will be aware of the logical mathematical property of fuzzy sets whereby 0.5 memberships create analytical difficulties. This is because it places the case in a logical limbo where verbal and formal logic are difficult to reconcile – the case is at the point of 'maximum ambiguity' and 'neither more in nor more out of the set'. The effect is that the coder has effectively decided that they cannot tell whether the case is a case of X or not X. The decision as to whether a case falls over or under this 'crossover point' in its set membership is a key consideration. It is normally a requirement to justify strongly the choice to choose, for example, 0.51 over 0.49, but I persevere for the sake of the example here. With this data the most significant consequences arise in analysing sufficiency relations, and are discussed in the next chapter.

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Table 6.2: 10-case fuzzy set membership scores

Case	Mayoral support	Citizen willingness to support contentious politics	Deepened democracy A	Deepened democracy B	Deepened democracy C
Belo Horizonte	0.51	1	0.51	0.67	0
Blumenau	0	0	0	0	0
Ipatinga	1	1	1	1	1
Porto Alegre (i)	0.51	1	0.51	0.67	0
Porto Alegre (ii)	1	1	1	1	1
Porto Alegre (iii)	0.51	1	0.51	0.67	0
Recife	0.51	1	0.51	0.67	0
Rio Claro	0	0	0	0	0
Santo André	0.51	0	0.51	0.33	0
São Paulo	0.51	0	0.51	0.33	0

and co-opted participatory democracy. It is important to note that in calibrating and coding the outcome I am calibrating a set score that Wampler has not explicitly derived. Wampler is clear that Porto Alegre 1997–2004 and Ipatinga achieve the quality of empowered participatory democracy and that there is no empowerment of note in Rio Claro and Blumenau (emasculated participatory democracy), so the appropriate set-membership for these cases is easier to interpret. As his purpose is ordering and categorizing, he never quite makes a distinction as to whether his two qualitative descriptions of 'informal and contested' and 'co-opted' participatory democracy display measurable differences in how far they achieve the outcome of empowered participatory democracy – only to say that the cases form a 'wide spectrum of outcomes' (2007a: 257).

Table 6.2 shows three different potential calibrations of both of these categories. The alternate calibrations of the outcome are used to show what is at stake in interpreting the work of other researchers. In particular, it allows a consideration of how a more systematic approach can highlight the extent to which parsimony in explanations of the kind provided in Table 6.1 is warranted. The six cases that do not have a set membership score of 1 or 0 are all more or less halfway in the set of 'deepened democracy A'. 'Deepened democracy B' shows informal and contested cases as 'more in' (0.67), while co-opted cases are 'more out' (0.33). Finally, I include 'deepened democracy C' as a calibration of the outcome

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to allow for a model where all cases are fully out except those in the empowered participatory democracy cell. Thus, I am focused on pointing out what is at stake in measurement, and showing that QCA can be used iteratively and transparently to elucidate and match measurement and theory.

Tests for necessity

The fundament of social science is knowing that information that reaches you is biased, and figuring out what level of uncertainty exists between evidence and theory. For necessity to be established, the set of cases containing the outcome must be a subset of the set of cases displaying the cause (Ragin, 2000: 214–17). Logically, for one condition or combination of conditions to be a subset of another condition, its membership value (the extent to which it is observed) should be less than or equal to the membership value of the superset across all cases. This is the relationship we are testing for, and that the claim of necessity or sufficiency is based on.[4]

I explained in Chapter 4 that researchers will usually seek to interpret conjunctions of conditions that pass a threshold of consistency or inclusion – 0.9 may be seen as a minimum consistency threshold for claims to necessary causation (Mendel and Ragin, 2011). In practice, researchers have usually accepted greater inconsistencies in evidence for a condition to be deemed sufficient than for necessary conditions. This is partly because necessary conditions are more often hypothesized and analysed in terms of single conditions, whereas sufficient conditions are more often evaluated as combinations of conditions. Single necessary conditions, by virtue of their presence being required in every instance of the outcome, impose more restrictions on where we might expect to observe an outcome in a way alternative sufficient combinations of conditions do not. Any number of conditions or combinations of conditions may be sufficient for a given outcome in different contexts. The assumption is that the same outcome can be reached in a number of different ways, and although some of these conditions can be more or less likely to occur, none of them is any 'better' at predicting the outcome when they do. There are, however, arguments from QCA scholars that some sufficient

[4] Ragin argues that these types of set-theoretic relations have been ignored by a focus on correlational patterns where such a relationship appears to suggest 'error' (Ragin, 2008).

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conditions may be 'empirically more important' than others on the basis that we have more evidence of their occurrence. And so, there are some other measures of fit that help understand those issues, which I explain if and when I draw on them.

When we run an analysis for necessary conditions based on the superset–subset relation across the cases, as coded in Table 6.2 with the outcome 'deepened democracy A', we get the output contained in Table 6.3.[5]

We find that in a necessity analysis of these 10 cases so operationalized, mayoral support is fully consistent with the necessity superset relation across the cases, indicating that it is a necessary condition for deepening democracy. Moreover, the negation of mayoral support is equally necessary to negate the deepening of democracy (this result is logical because their set membership values are equal in every case). That is, mayoral support alone is necessary and sufficient for deepened democracy based on that data.

CSO willingness to use contentious politics does not seem to be necessary for deepened democracy on this account, contrary to Wampler's claim. Why? When we examine the cases, we see that this is because both Santo André and São Paulo show some degree of deepening democracy in this calibration. In justifying the coding, I might say that on my interpretation, Wampler acknowledges this 'medium quality' of deeper democracy when he makes the qualitative distinction between co-opted democracy, where 'citizenship rights and accountability were weakly extended' (2007a: 214) and emasculated democracy. Therefore, cases that appear to have some relevant deepening, yet low or no evidence of CSO willingness to use contentious politics, do not confirm the necessary superset–subset relationship.[6]

'Deepened democracy B' offers another interpretation of deepening democracy where Wampler's co-opted cases are more out of the set

[5] Analysis performed using the superSubset command in the QCA package for R (Dusa, 2019) and xy.plot command in the SetMethods package (Oana and Schneider, 2018). Throughout I adopt the common notation where the tilde '~' preceding the letter denotes absence (negation) of a condition; the asterisk '*' denotes intersection of sets (conjunction of conditions); and '+' denotes union of sets (disjunction). Data and replication code for all the analysis presented in this book is available at the University of Southampton Institutional Repository: https://eprints.soton.ac.uk/444081/

[6] There also seems to be evidence of some contentious politics in the cases, but it was not of a successful kind. This is reflected better in the larger-N analysis and serves to highlight the trade-offs in parsimony and complexity in explanations.

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Table 6.3: Analysis of necessary conditions for 10 cases for outcome 'deepened democracy A'

Analysis of necessary conditions outcome variable: deepened democracy A			
Conditions tested	Inclusion	Coverage	Relevance
Mayoral support	1	1	1
~mayoral support	0.581	0.595	0.717
Citizen willingness to use contentious politics	0.798	0.673	0.671
~citizen willingness to use contentious politics	0.202	0.255	0.668
Analysis of necessary conditions outcome variable: ~deepened democracy A			
Conditions tested	Inclusion	Coverage	Relevance
Mayoral support	0.595	0.581	0.7
~mayoral support	1	1	1
Citizen willingness to use contentious politics	0.397	0.327	0.498
~citizen willingness to use contentious politics	0.603	0.745	0.855

of deepened democracy, but the informal and contested cases are more in the set, even if they are not fully in. The calibration of the outcome set requires the researcher to tackle the question directly as to whether the variation in democratic deepening among unempowered cases is relevant to explaining democratic deepening overall. On the crisp 'C' calibration, these middle cases are all excluded from deep democracy – on the 'C' interpretation partial delegation of authority to citizens is not enough to constitute deep democracy. Wampler suggests that the middle cases show some delegation of authority, but that this is hampered by contradictory mayoral strategies (2007a). Whether any observer wishes to allow the conceptual expansion of what deepened democracy constitutes to include more of these cases will remain always, to an extent, a judgement based in democratic theory with some normative content. What QCA allows us to do is better understand the implications of the judgement for claims aimed at causal explanation.

There is some inconsistency in all necessity inclusion scores if the outcome 'deepened democracy' is represented by the set 'deepened democracy B'.[7] With it the argument for any necessary conditions becomes less plausible. For both conditions we have cases that contradict the necessity relationship (consistency or inclusion score for mayoral

[7] Not shown in the tables.

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support would be 0.88, for citizen willingness to support contentious politics, this would be 0.876). The uncertainty regarding the necessity of mayoral support is introduced because with 'deepened democracy B', some cases now have greater membership in the set representing the outcome than they do in the set representing the explanatory condition, as shown in Figure 6.1. As social scientists willing to admit some uncertainty in our explanations, we might be happy with a relatively low level of inconsistency across all the cases and make generalizations with transparent reflection on this interval of confidence. As explained earlier, however, necessary conditions set a particularly high standard for causal explanation, and 0.9 may be seen as a minimum consistency threshold for claims to necessary causation (Mendel and Ragin, 2011).

The only calibration that confirms Wampler's inference that both conditions are necessary for deepened democracy is C (see Table 6.4). This is probably also the fairest interpretation of Wampler's work. This calibration sets apart the distinct category he calls institutionalized participatory democracy. Note that the necessity claim requires a conceptualization of 'strong participatory democracy' or 'deepened democracy' that excludes even partial membership to all cases of informal and contested or co-opted participatory democracy. If mayoral support and citizen contentious politics conditions are necessary, it is because our conceptualization of deepened democracy assumes that the differences between co-opted, informal and contested and emasculated participatory democracy are irrelevant to explaining democratic deepening. For purists, this interpretation might make a lot of sense. For others, it might be too restrictive.

The latter position is supported by the measures of coverage and relevance that readers can notice have been lowered significantly for the conditions that are consistently necessary in Table 6.4 compared to Table 6.3. What does this mean, and how should it be interpreted? Coverage and relevance of necessity are both measures of the triviality of necessary conditions (Schneider and Wagemann, 2012: 147; see also Goertz, 2006).[8] The 'C' calibration is restrictive and therefore deepened democracy becomes a much smaller set of elements relative to the rest of the cases within the scope of investigation – only 2 out of our 10 cases display the outcome and the rest do not. This skew means that

[8] Necessity coverage is an expression of how much smaller the outcome set is in relation to the conditions set (how much of Y covers X). Low necessity coverage can indicate trivialness of necessary conditions (X is quite big relative to Y), and high coverage can indicate a relevant necessary condition, provided X is not too close to a constant (see Schneider and Wagemann, 2012: 232–7).

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Figure 6.1: Necessity of mayoral support for 'deepened democracy B'

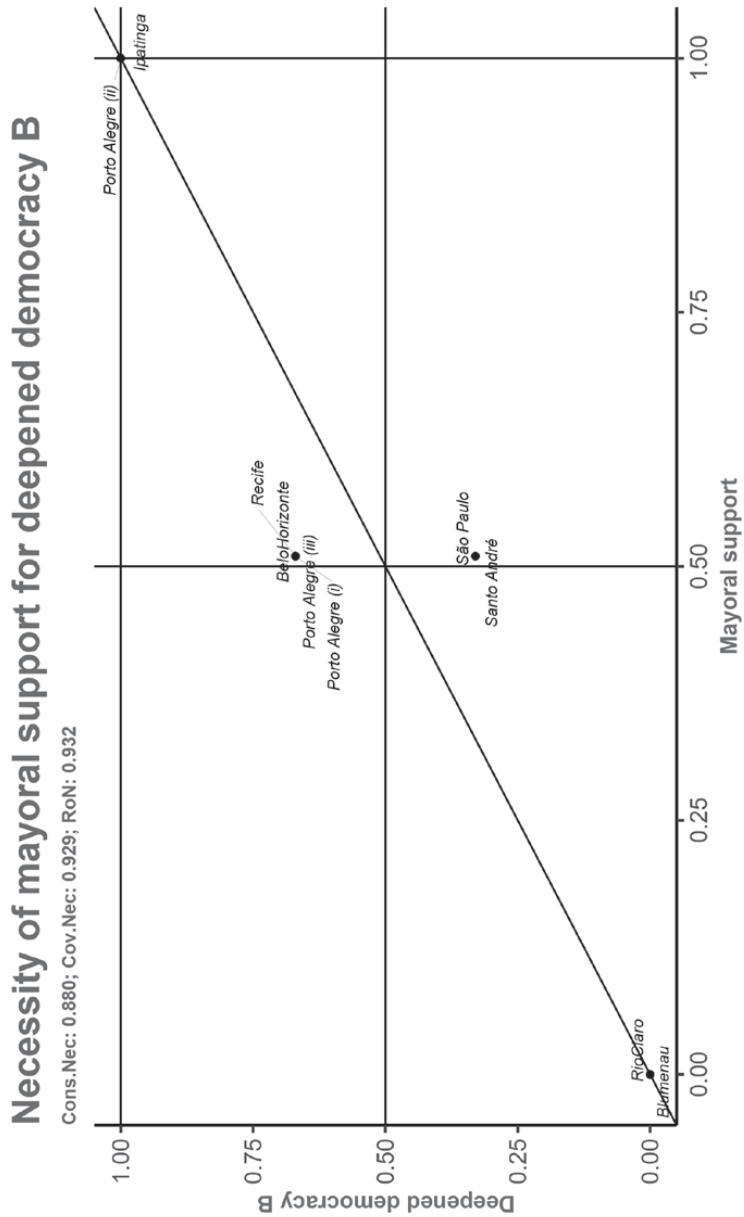

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Table 6.4: Analysis of necessary conditions for 10 cases for outcome 'deepened democracy C'

Analysis of necessary conditions outcome variable: deepened democracy			
Conditions tested	Inclusion	Coverage	Relevance
Mayoral support	1	0.395	0.618
~mayoral support	0	0	0.506
Citizen willingness to use contentious politics	1	0.333	0.5
~ citizen willingness to use contentious politics	0	0	0.6
Analysis of necessary conditions outcome variable: ~deepened democracy			
Conditions tested	Inclusion	Coverage	Relevance
Mayoral support	0.382	0.605	0.712
~mayoral support	0.618	1	1
Citizen willingness to use contentious politics	0.5	0.667	0.667
~ citizen willingness to use contentious politics	0.5	1	1

it is much easier for an outcome to be a subset of causes (a necessary condition). The logic exposed is that more restrictive conceptualization of democracy will set an ever-lower bar for necessity and a higher one for sufficiency such that almost anything can be seen as necessary and almost nothing is sufficient. One can easily imagine how such a finding might expose over-conscious thinking and be deemed a trivial or less relevant finding by a practitioner looking for information on how to act to deepen democracy in a practical context.

The point of this short example is to show that the act of careful calibration of our concepts to represent the evidence and tests for superset relations can reveal something to us about our assumptions. The most important criterion for any conceptual description or measurement of the social world is that they reflect the world as accurately as possible. QCA is a tool that can entreat us to better specify what we mean when we talk about, for instance, 'success' in democratic innovations, which many have suggested is a term that is often used uncritically (Montambeault, 2016; Spada and Ryan, 2017).

A QCA approach and analytic test can provide a useful logical check on the congruence between evidence, intuition and measurement. Most analytic approaches in social science, however, are not content with stopping at the level of describing categories – their goal is to infer. If concepts (descriptions of qualities) are sets of which empirical cases have

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degrees of observed membership (fuzzy membership or quantitative variation), the relative degree of membership matters greatly if we want to engage in causal analysis – in this case, assessing claims of necessity and sufficiency. QCA maps the distance between the theory and the evidence, and allows transparent iteration and re-evaluation. We may conclude that QCA as an approach has added some value to the analysis in this first instance by problematizing our conceptions of democratic deepening, and forcing us to think about what substantive conclusions on causation of the phenomenon are suitable. The analysis suggests that variation in the use of contentious politics means that it can only be deemed a necessary condition if we prefer a very restrictive definition of deepened democracy, and although most readings are likely to find participatory leadership a necessary condition, that summary doesn't completely evade question based on one reasonable interpretation of the evidence. We can probably surmise that any conclusions based on a couple of conditions elide too much information, and we need to include more for accuracy.

Extending the analysis: bringing complexity back in

This discussion is only based on a stylization of Wampler's most parsimonious conclusions and not explicitly on delving fully into his richer descriptions of the cases. While small-N research often concludes by extracting what the researcher feels are the two or three most important explanatory conditions, it can also claim advantage in its rich case description – elucidating the trail of relations and processes involving other or secondary conditions. In some large-N statistical analysis these details may be overlooked even in assessing interactions, as the compound variable represented by the interaction term is often unrecognizable from its elemental beginnings.

Wampler details any number of potential influencing conditions or variables from which other hypotheses may be formed and tested to investigate causal chains relating to deepening democracy. These include population size, social backgrounds of PB delegates, higher human development index (HDI) scores, and so on. At one point he suggests a conjunctural hypothesis whereby cities with greater HDI are more likely to be left-leaning, have broad-based CSO activity and a firmer financial base on which to administer PB, advising that they may play the role of mediating variables. In his chapters describing each case, however, he highlights and details, in particular, mayor–legislative relations, the financial basis for implementation of the programme and the rules of the game as influential in determining the depth of

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democracy coming out of PB processes. In fact, he affords PB rules the same explanatory primacy as CSO activity and mayoral support, although as we shall see, operationalizing it as a variable in comparative analysis is a little tricky, which is possibly why he does not highlight it as much as the latter two in Table 6.1. In short, although Wampler searches for an appropriate parsimony, his work, like that of many others discussed in this book, takes the likely complexity of causality seriously.

The key strength of traditional qualitative small-N research approaches vis-à-vis cross-sections or QCA, is that it is not bound by a 'static' analytic moment and allows fluidity in causal tracing. At the same time, this is also its weakness. A case researcher submerged in subjective determination of causal processes is in some important ways less well placed to make hands-off 'deductive' judgements about how variables combine. Wampler, in his excellent study of PB, may suggest that based on his case knowledge, these other three conditions (mayor–legislative relations, the financial basis for implementation of the programme and the rules of the game) are important but secondary in the story to the two first discussed. But in a QCA approach we can transparently analyse all possible combinations of the five conditions and potential counterfactual assumptions on an equal footing.

An improved set calibration for a more deductive analysis?

In this section I briefly explain Wampler's conditions and provide details of how sets of conditions are calibrated for an enlarged analysis. The section draws heavily on Wampler's theoretical justifications for highlighting these conditions. His evidence is particularly useful because he often triangulates evidence from interviews (that is, subjective determinations of, say, mayoral support) with symptomatic indicators of degrees to which a condition is observed (for example, in the case of mayoral support, implementation rates of PB projects and their prioritization vis-à-vis projects decided on through other channels). This analysis showcases some of the deductive features of QCA before I present the worldwide analysis of several cases drawing on accumulated evidence from several studies.[9]

[9] I keep the calibration discussion somewhat brief for this analysis of one study in recognition that I have described in detail the ultimate comparison of PB worldwide in Chapter 5. The main purpose here is to show how QCA can have some advantages for theory evaluation even when working with a smaller-N.

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The outcome of interest, 'deep democracy', is seen where PB programmes result in open, equal and meaningful participation of large numbers of 'ordinary citizens', and where the PB overtakes previous clientelist practices as the main method for citizens and civil society to negotiate and realize budget priorities. I do not take the most restrictive approach to conceptualization here. The cases above the crossover point on this interpretation are those where Wampler reports relatively significant delegation of authority leading to implementation of projects and high participation rates (even where results came via struggle). I give lower set-membership scores to co-opted cases where delegation of authority was more an appearance and cherry-picking of projects from PB was the norm. My interpretation is that this is a justified claim and can be read through the case descriptions. Also, the experience of co-option is more likely to reduce support for democracy and feelings of efficacy vis-à-vis the experience of contestation, which, while it frustrates citizens, can lead to some empowerment through instilling forms of efficacy over time.

In explaining how PB programmes can achieve deep democracy, Wampler puts forward five key causal conditions. The first is 'high mayoral support for PB'. Mayoral support is vital because decision-makers 'must be willing to spend scarce resources' (Wampler, 2007a: 36). There may be instrumental reasons for high mayoral support of PB programmes (as a signalling device to gauge citizen preferences, or as a political party support-building measure) as well as ideological. In the Brazilian context mayoral support is taken to be key to outcomes because strong mayoral support can lead to the implementation of projects, allowing demonstration effects (see Abers, 1998; Gret and Sintomer, 2005: 87). The argument goes that when governments are seen to be taking the process seriously by implementing the decisions of the participants in a participatory process (in particular, building capital infrastructure projects), the institutionalization of PB increases. In turn, this can give greater scope to governments to reorganize bureaucracy to administer PB constructively.

The second explanatory condition set is that of strong civil society using contentious and cooperative politics. Strong civil society can cooperate with other actors in deliberative forums, but also contest information vigorously inside and outside fora, defending rights using contentious forms of political action where required (Wampler, 2007a: 38). The types of activities CSOs are willing to

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engage in can often be explained by the historical density of CSOs in a municipality.[10]

For various situational and political reasons rules vary from one PB programme to another. We can talk of the set of 'rules that delegate authority' where rules allow citizens accountable and direct decision-making, which can incentivize greater and more meaningful participation. According to Wampler, 'the unintended consequence of unclear rules is a limited delegation of authority' (2007a: 39). Combining rules in an explanatory model with the role of actors using QCA (which explicitly investigates conjunctural causation) is important if we believe that institutions influence actors and actors influence institutions.

Wampler emphasizes throughout his book the combinatorial effects of conditions including how one condition may limit the degree to which any combination can be effective in deepening democracy through participation. This is another reason why the analysis may lend itself to QCA and fuzzy sets. One example is the extent to which a mayor's strategy is conditioned by the existence of a 'positive legislative environment'. The legislative environment is less favourable to an outcome of deeply democratic PB when the mayor implementing the programme does not have a broad base of support and must spend political capital shoring this up (2007a: 40). As we saw in Chapter 2, the strength or weakness of opposition and importance of broad-based political support for PB is emphasized in numerous accounts. Wampler suggests that this positive environment is a necessary but not sufficient condition for deeply democratic PB – the mayor may still not wish to delegate authority even with legislative support, but will find it difficult to delegate without. Wampler's descriptions also suggest that this condition could highlight asymmetric causation when he outlines that in the case of Santo André, a highly supportive legislature can actually incentivize the mayor to engage in many other projects, undermining the importance of PB (2007a: 209).

Finally, the 'financial basis for spending', that is, the availability of significant funds for new capital investment, is also, Wampler holds, necessary but not sufficient for PB to work effectively. This is because

[10] Note that this condition itself can be considered a conjunction – a logical combination of contentious politics and cooperative politics in CSOs. The condition representing the combination could be constructed by calibrating separately both conditions and calculating the intersection of both sets. In the example I do this implicitly (that is, in calibrating I do not compromise high contentious politics for low cooperation, and vice versa).

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limited spending ability limits the power of programmes where authority is delegated to citizens.

After enumerating these conditions and explanatory hypotheses, Wampler contends that his explanations of cases will show that 'it is necessary for a PB program to have positive results in each area to produce a successful PB program' (2007a: 41). Note that these claims add a higher bar to the parsimonious claims that make up the typology discussed in the earlier example. I aim to test here Wampler's more complex claim, basing a calibration of conditions on the narrative he provides, and using fsQCA.

I calibrate sets and ascribe case membership in each case. A good starting point is to outline the verbal meanings we ascribe to set membership. As per Chapter 5, the following 7-value fuzzy set, as proposed by Ragin (2000: 156), is used.

1.0	Fully in (the set)
0.83	Mostly but not fully in
0.67	More or less in
0.5	Crossover point
0.33	More or less out
0.17	Mostly but not fully out
0	Fully out

This seems a feasible level of nuance to justifiably extract from the qualitative information in Wampler's book. He provides at least a few hundred words (and often a lot more) of descriptive information on each condition for each case. This rich description is necessary for an epistemologically sound QCA.

The data matrix presented in Table 6.5 discloses my calibration of the conditions.[11] I provide my most rigorous interpretation, but it is an interpretation of the work of others. While I might claim to 'control for the researcher' in my application of QCA to Wampler's cases (which I won't do when cumulating across researchers), I cannot make any exceptional claims to being able to control the ever-present problems for researchers interpreting one another's descriptions and findings. One advantage of

[11] I omit the post-2004 Porto Alegre case here because there is not the same level of description of it in this source material. For robustness I also considered the effect of slight changes to fuzzy memberships in order to help decide on appropriate consistency thresholds. Tests showed no effect on the ordering of cases within sets.

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working with fuzzy sets is that it can at least lay those interpretations out in a systematic way for others to query, rather than hide them in implicit assumptions.

There are some changes from the earlier calibration based on my reading of the 'thicker' case descriptions. For instance, São Paulo and Ipatinga's scores for civil society using contentious and cooperative politics are on opposite sides of the crossover point to the data matrix in the first analysis described earlier. The introduction of more complexity can regain some anomalies that are not clear in more parsimonious conclusions. Building on the first part of the chapter, which aimed to show what is at stake in different calibrations and selecting levels of measurement, this part aims to investigate what is at stake in more parsimonious or complex causal modelling when employing case-based comparative logic.

As these measures are based on my interpretation of text and not on a ranking systematized by Wampler, as in the earlier analysis, it is also pertinent to make some comments on the difficulties involved in calibrating conditions to allow for transparency for those wishing to repeat the analysis. The rules-based condition was particularly difficult to calibrate because it requires some subjective interpretation of how even small changes in basic rules may affect participant strategies. This is often tricky to separate from outcomes in the process of PB. As rules are, in the Brazilian case, often set by the mayor, they may be better conceived of as a symptom of that support. The financial basis condition also requires decisions on how to weigh absolute and relative financial strength and include issues like inheritance of debt. By presenting these as a data matrix in Table 6.5 they can be open to scrutiny by others with knowledge of the case.

Necessity

I am particularly interested in Wampler's claim that all five conditions identified are necessary for democratic deepening, and that a positive legislative environment and financial basis for support of PB are necessary, but not sufficient (2007a: 35). This number of necessary conditions seems a high bar. Also, if we think these latter two conditions are necessary but not sufficient, then which combinations of conditions are they necessary components of? Without any tool to analyse the consistency of this claim, case researchers are disincentivized from problematizing the combinatorial relationships between variables.

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Table 6.5: Data matrix for Wampler's cases (5 causal conditions)

Case	**ms**	**ccp**	**ple**	**fbs**	**rep**	**deepd**
Belo Horizonte	0.33	0.83	0.67	0.83	0.33	0.67
Blumenau	0.17	0.33	0.17	1	0.67	0
Ipatinga	0.83	0.33	0.67	1	0.67	0.83
Porto Alegre 1988–96	1	1	0.83	0.33	1	0.83
Porto Alegre 1997–04	1	1	0.67	1	1	1
Recife	0.67	1	0.67	0.17	0.33	0.67
Rio Claro	0.17	0	0.17	0.67	0.17	0
Santo André	0.83	0.83	1	0.33	0.33	0.33
São Paulo	0.17	0.83	0.33	0.33	0.33	0.33

Note: ms is mayoral support; ccp is civil society using contentious and cooperative politics; ple is positive legislative environment; fbs is financial basis for spending; rep is rules that encourage participation and deepd is deepened democracy.

The consistency of single necessary conditions for the outcome 'deep democracy' and its negation (not deep democracy) is reported in Table 6.6.

Consistency of the necessary superset relations for individual conditions range in value for each of the five hypothesized causal conditions for deepened democracy, from 0.79 to 0.93.[12] In other words, many of these conditions come close to the criteria for construing causal necessity (an almost always necessary condition), but all suffer from some inconsistencies. A high threshold of consistency to make claims about necessity is advisable (at least 0.9). However, as Ragin, among others, repeatedly stresses, a further important test is whether it 'makes sense' based on knowledge of theory and background conditions as a necessary condition (Ragin, 2000; Mendel and Ragin, 2011). There are a number of strategies the researcher now has open, the consequences of which they choose will affect the interpretations not just of necessity but also of sufficient conditions.

As many of these conditions could be seen as 'almost' necessary, we now have an opportunity to revisit each condition and investigate how the subset relationship is contravened. One contravening

[12] Rounded to two decimal points. As no condition in the analysis of the negation of the outcome (absence of deepened democracy) is highly consistent with a necessity subset–superset relation, I only discuss the analysis of the outcome (deepened democracy) in this section.

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Table 6.6: Superset relations for single necessary conditions (5 condition analysis)

Analysis of necessary conditions outcome variable: deepened democracy			
Conditions tested	Inclusion	Relevance	Coverage
Mayoral support	0.927	0.784	0.811
~mayoral support	0.358	0.727	0.455
Civil society using contentious and cooperative politics	0.893	0.589	0.676
~civil society using contentious and cooperative politics	0.253	0.786	0.414
A positive legislative environment	0.895	0.791	0.805
~ a positive legislative environment	0.391	0.721	0.476
The financial basis to spend	0.785	0.625	0.647
~the financial basis to spend	0.466	0.829	0.65
Rules encouraging participation	0.82	0.805	0.791
~rules encouraging participation	0.5	0.724	0.559
Analysis of necessary conditions outcome variable: ~deepened democracy			
Conditions tested	Inclusion	Relevance	Coverage
Mayoral support	0.539	0.551	0.439
~mayoral support	0.767	0.94	0.907
Civil society using contentious and cooperative politics	0.615	0.45	0.434
~civil society using contentious and cooperative politics	0.541	0.925	0.825
A positive legislative environment	0.539	0.574	0.452
~ a positive legislative environment	0.767	0.914	0.872
The financial basis to spend	0.730	0.573	0.560
~the financial basis to spend	0.539	0.85	0.701
Rules encouraging participation	0.576	0.642	0.518
~rules encouraging participation	0.767	0.852	0.799

fuzzy membership value in a case may require reconsideration. For example, the condition relating to civil society only contravenes the necessity superset relationship in Ipatinga. Figure 6.2 shows it is the only case lying above the diagonal and having greater membership in deepened democracy than in citizen support for contentious politics. Can we go back and explain the deviant case? Is there a coding error? I have coded the case as more out than in the set of citizens using

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contentious and cooperative politics. The description of the case reports what contentious politics exists as being disagreement only within the confines of PB and only by organizations that are allies of the mayor. One could contest the interpretation. Alternatively, perhaps, if we look at the other cases in this sample, those that achieve deepened democracy appear to be large provincial capitals, and this is likely to affect the nature of civil society in the case. Therefore, one could make the argument, revisiting the case data, that for the purposes of modest generalization we could drop some cases from this particular claim and limit the scope of the generalization on the necessity of willingness to use cooperation and contention to larger cities only.

Shifting the scope of an argument in response to findings requires careful consideration. It can be all too tempting for a researcher to move away from the logic of the initial case selection to find an answer that fits the theory. In particular, for the interpretation of necessity we should set a high standard because what we interpret to be necessary has implications for what combinations of conditions we will accept as sufficient, and thereby for the theories of change we try to implement in the world.

A second consideration in seeking to establish evidence for a necessary condition would be to consider disjunctions of conditions (Bol and Luppi, 2013). We can ask which unions of two or more conditions (disjunctions) are consistent with a necessity subset–superset relation. Table 6.7 shows all the unions of two conditions with necessity consistency over 0.95. It is not uncommon to see proclamations in the PB literature that either this *or* that is necessary for an outcome. From the nine disjunctions listed in Table 6.7 we can see, for instance, evidence that either a civil society willing to engage in both contentious and cooperative politics OR a secure financial basis for spending on projects are present when we observe deepened democracy in PB programmes. Either one or the other is necessary. These can be called SUIN conditions – a Sufficient but Unnecessary part of a factor that is Insufficient but Necessary for an outcome (Mahoney et al, 2009). A researcher can say with a greater degree of certainty based on the QCA necessity analysis below that deep democracy cannot be achieved in PB programmes without at the very least a financial basis to spend OR civil society willingness to both struggle and cooperate with government. The interpretation of SUIN conditions is tricky, however, because in the same way that I discussed the perils of restrictive theories earlier, a theory that adds a disjunct logically lowers the barrier for evidence of its veracity by providing new avenues for cases to meet

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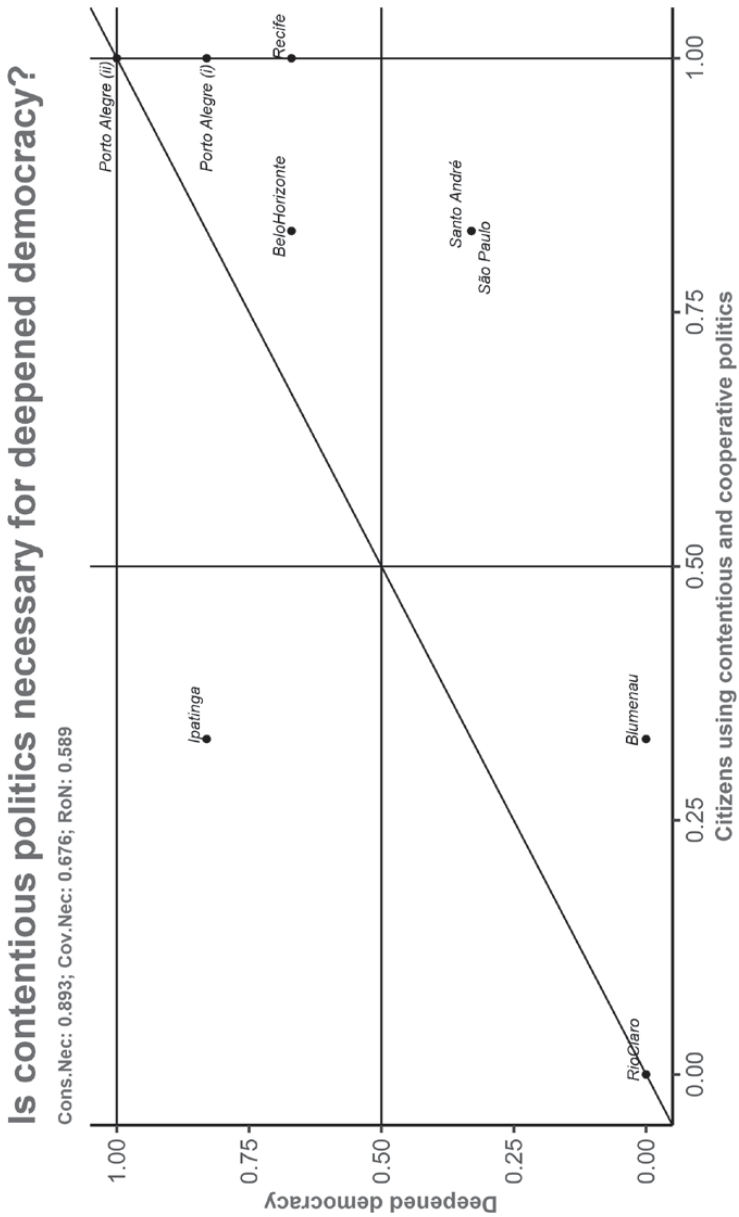

Figure 6.2: X–Y plot of set membership in deepened democracy and civil society support for contentious and cooperative politics

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Table 6.7: Necessity consistency for Boolean sum expressions

Boolean expression	Necessity consistency	Relevance of necessity	Necessity coverage
MS + CCP[1]	1	0.502	0.683
MS + PLE	1	0.728	0.798
MS + FBS	1	0.385	0.636
MS + ~rep	1	0.5	0.682
CCP + PLE	0.966	0.482	0.659
CCP + FBS	1	0.194	0.571
CCP + REP	0.966	0.444	0.643
PLE + FBS	1	0.385	0.636
PLE + REP	0.966	0.627	0.728

[1] In QCA notation '+' signifies logical OR (the substitutability of two conditions) while '*' signifies logical AND (the combination of two conditions).

the required conditions. We would need good theoretical grounds to suggest that two conditions are practically substitutable to produce an outcome and represent some higher-order construct. I hope to have therefore exposed the many potential interpretations of necessity and the trade-offs that researchers should consider in making causal claims of necessity.

I hope readers will not see such exhaustive analysis as introducing unnecessary complexity. It could be seen as a terrific example of the dangers of over-conscious thinking and methodological fetishism. But my goal here is to show just what is at stake in the underlying assumptions when researchers state that X is a necessary condition for Y, and I will show that models espousing many necessary conditions will have strong implications for hypotheses about what combinations of conditions structure success in PB, and what practical action we can take to achieve success.

In terms of added value, we can at the very least say that the necessity analysis here has cast some doubt over Wampler's general contention (within the scope of his population) that all five conditions are necessary for deep democracy. Perhaps our judgement is clouded because the evidence for relationships does exist, even if they are not determining across time and space, and many of these factors can be considered substitutable necessary conditions. This adds nuance to the general claim. Is the inconsistency in claims for single necessary conditions mirrored across all the cases in our main dataset?

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Cumulating research: necessity in participatory budgeting across several cases

After introducing the analytic process as it is applied to cases that have previously been analysed together, I now turn to analyse the accumulated research of the 31 cases of PB presented in Part II of the book. Recall that these cases cover a range of studies by different researchers, and I am interested in explaining citizen control of collective decisions in PB by examining combinations of political leadership, bureaucratic support for programmes, civil society demands and the financial basis for support for programmes (fuzzy set scores outlined in Table 5.1 in Chapter 5). Are there, as has regularly been claimed in the literature, several necessary conditions including among them participatory leadership and civil society demand, for citizens to gain control of budget decisions? Table 6.8 presents all single conditions and disjunctions of two conditions that have a score of necessity inclusion that is greater than 0.9 across all 31 cases.

None of these conditions perfectly fits the necessity relationship. A political leadership committed to participation has long been considered a necessary condition for success in producing empowering programmes. The condition receives an inclusion score of 0.93 across the cases, indicating evidence of necessity. It is worth investigating further. The X–Y plot in Figure 6.3 visualizes this relationship.

Overall the evidence that participatory leadership is a necessary condition for citizen control is quite strong. The cases below the diagonal are fully consistent with necessity (X>Y). The four cases that contribute to inconsistency, placing doubt in the claim, are Toronto, Buenos Aires, Cusco and Recife (ii), that is, Recife under the PT. Going back to investigate these specific cases offers some nuance to the

Table 6.8: Necessary conditions for citizen control of PB with consistency greater than 0.9

	Conditions	Inclusion	Relevance	Coverage
1	PL	0.933	0.652	0.698
2	BSP + CSD	0.91	0.637	0.68
3	BSP + FBS	0.911	0.629	0.675
4	CSD + FBS	0.91	0.572	0.643

Note: PL is participatory leadership; CSD is autonomous civil society demand; BSP is bureaucratic support; and FBS is financial basis to spend.

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necessity claim. None of these cases are what Schneider and Rohlfing refer to as 'deviant cases in kind' (2013: 581). That is, we do not have any examples in the upper left corner of the plot of strongly deviant cases that clearly achieve successful citizen control while having very little evidence of participatory leadership. Cases like Toronto and Buenos Aires might normally be considered less relevant because they are neither comparatively good examples of citizen control nor of strongly committed participatory leadership (for a discussion of how to interpret such cases, see Haesebrouck, 2015). But on this occasion they are quite instructive.

Toronto's community housing PB is a case whose membership among successful cases has been changed from the pilot studies in which I previously presented it (Ryan and Smith, 2012, 2014). This is because more recent research suggests far more mixed outcomes in the case (Foroughi, 2017). Nevertheless, early accounts of the case provide clear evidence for citizens having substantial influence over major spending decisions. Lerner and van Wagner suggested that absence of participatory leadership contributed positively to improved outcomes: 'participatory budgeting emerged when staff were passionate and prepared' and 'politicians were looking the other way' (2006: 15). The idea has some affinity with Dennis Rodgers' description of the Buenos Aires case (2010). There, politicians reeling from the *Argenitazo*[13] initiated a top-down PB, with mostly short-term instrumental goals in mind. Even when political support was abandoned as political manoeuvres within the governing coalition soon acted to undermine the PB, committed bureaucrats were sometimes able to use their skills to counteract those trying to subvert the process and allow citizens to retain decision-making power and influence spending (Rodgers, 2010: 20). The cases suggest that some democratic deepening can occur in the short term without participatory leadership, but the question remains as to whether it can be sustained.

This direction of travel seems to be confirmed by the other more successful cases where there is some degree of deviance in necessity inclusion. McNulty considers Cusco to be the most successful of the cases she studied in Peru (2011: 114), but here, also, support from the political leadership was not immediately clear. McNulty explains that the regional president won with a small margin of victory, faced strong opposition, and his behaviour did not suggest he was a fan of citizen consultation (2011: 111). Nevertheless, a combative civil society led to an accommodation of demands that seemed to later convince the leadership that participatory strategies would serve their interest. The

[13] See https://www.theguardian.com/world/2003/jan/25/argentina.weekend7.

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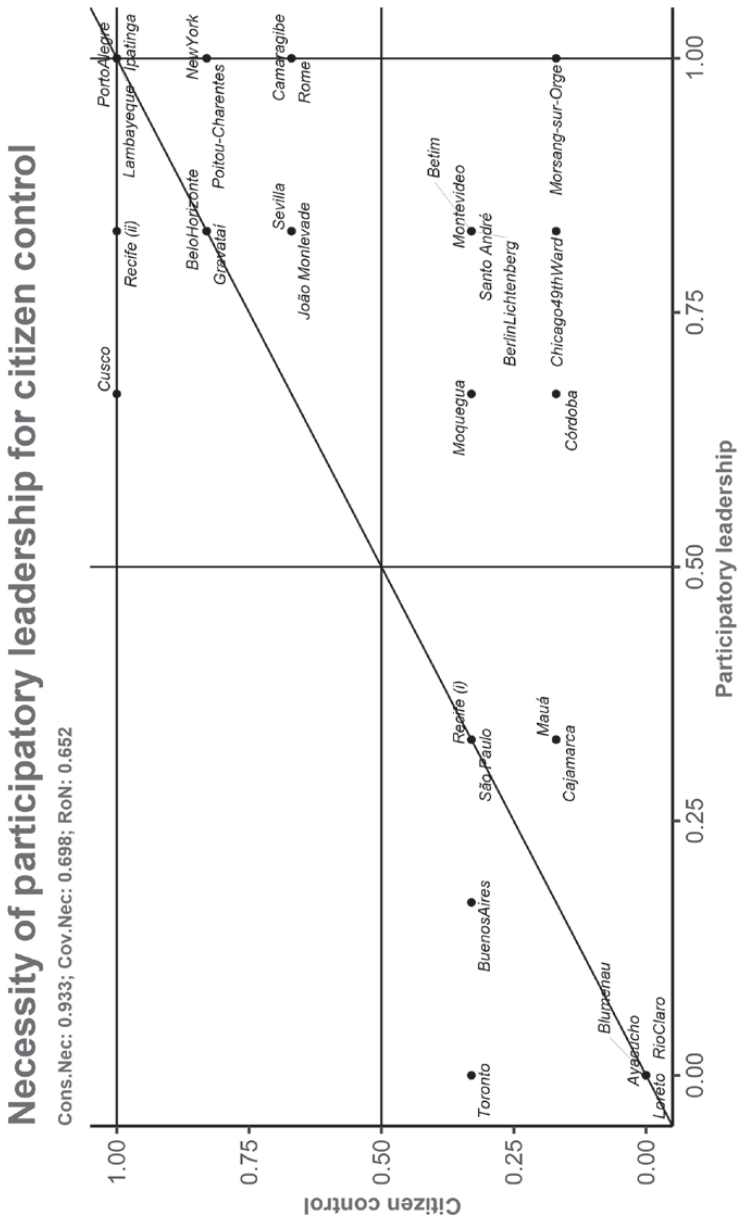
Figure 6.3: Necessity of participatory leadership for citizen control



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Recife (ii) case is observed to have slightly lower than full membership in the participatory leadership condition not because there is any question in the PT mayor's commitment to PB, but because he faced significant opposition to PB from within his coalition and outside it. The administration was able to ride that wave, and PB's success in Recife seems to have reached its zenith later in the decade.

This dive back into the cases on being presented with comparative evidence then suggests some nuance to the claims regarding participatory political leadership being a necessary condition for PB to achieve success in improving democracy that has been presented up to now. Participatory leadership is not a necessary condition for democratic deepening in the short run. In some cases, we can see that programmes cope with leadership absence where other actors are mobilized, sometimes providing democratic goods without political leadership for a time. But it is difficult to see any evidence of long-term good outcomes absent participatory leadership (although such cases might still be imagined). Participatory leadership comes closest to satisfying the test for a necessary condition. There is some remaining doubt, but I have shown that those issues appear to be explicable and with some nuance added, the claim gains further credibility. Future investigations of cases may be useful to test the hypothesis that a lack of supportive participatory leadership can be endured, but only for a couple of years.

On the whole, however, it is best to consider the several claims about necessary conditions for democratic outcomes in such innovations to have been somewhat premature. In early accounts the civil society condition was often held up as a necessary condition. This thesis was hardly questioned by first observers of PB. However, all conditions apart from participatory leadership show significant inconsistencies in tests for necessity. The plot in Figure 6.4 shows several pointedly deviant cases for that relationship. In some of the cases the deviance reflects successful cases where civil society was mobilized in a partisan way, but in others civil society was almost absent from the process in successful cases. The same pattern can be observed for the other conditions tested. There is a reasonably compelling amount of evidence to suggest participatory budgets are successful in providing meaningful citizen control in instances where spending is relatively constrained and where officials see the programme as a burden and fail to support it.

What of necessary conditions for the absence of citizen control of decisions in PB? Again, there is no evidence for such a strong claim, as shown in Table 6.9.

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Figure 6.4: Fuzzy X–Y plot for necessity of citizen control: autonomous civil society demand

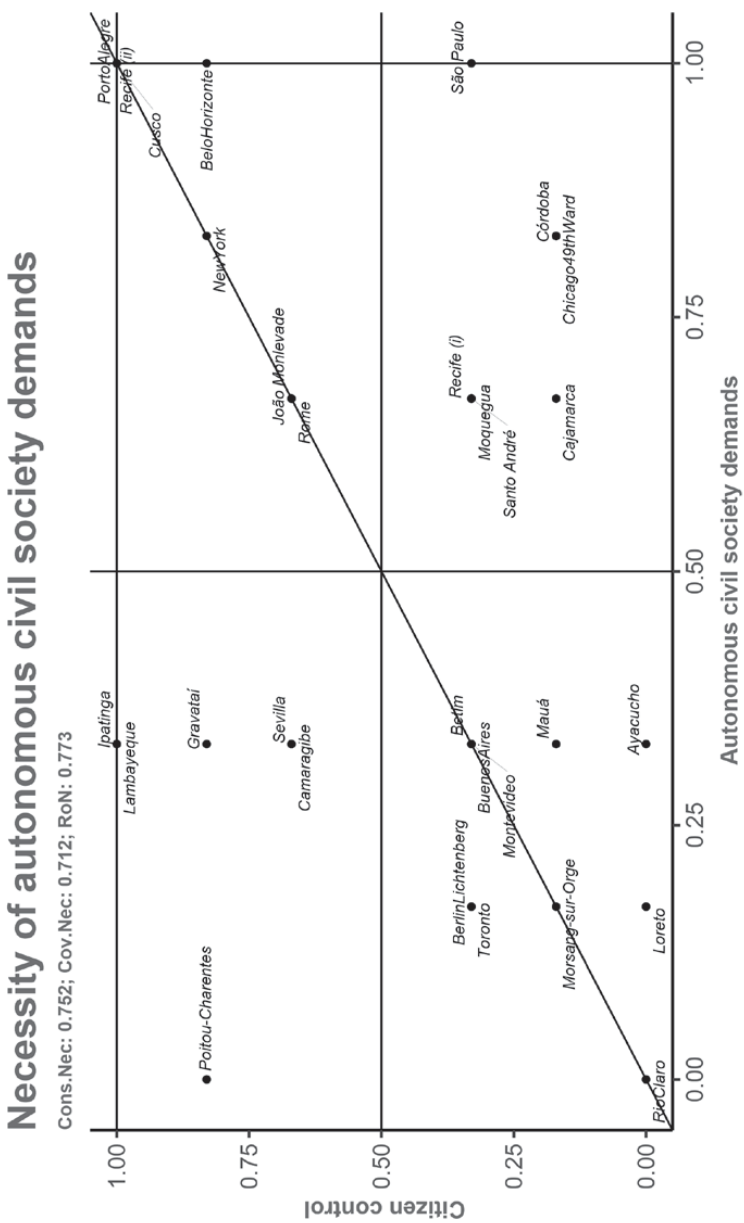

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Table 6.9: Test for necessary conditions for the absence of citizen control of PB decisions

	Condition	Inclusion	Relevance	Coverage
1	~participatory leadership	0.631	0.952	0.912
2	Participatory leadership	0.567	0.513	0.463
3	~bureaucratic support	0.845	0.76	0.767
4	Bureaucratic support	0.413	0.733	0.507
5	~civil society demand	0.721	0.81	0.761
6	Civil society demand	0.577	0.708	0.596
7	~financial basis to spend	0.702	0.772	0.718
8	Financial basis to spend	0.517	0.699	0.551
9	~csd + ~bsp	0.959	0.58	0.705

Note: csd is autonomous civil society demands and bsp is bureaucratic support.

One disjunction of two conditions (shown in the final row of Table 6.9) reaches greater than 0.95 consistency – that is, the absence of vibrant civil society demand-making OR the absence of bureaucratic support. It might be that one could interpret this as a useful finding if willing to consider it evidence of a higher-order construct such as 'absence of human capital', but this feels a bit of a stretch without greater recourse to theory and empirical evidence. I argue that the approach I have taken advances our theoretical understanding and provides clear signs for where future investigations should focus their interest to reduce further uncertainties.

In summary, it is better to say that the evidence presented in this chapter shows that the majority of claims, and perhaps all strong claims to necessity in PB research, have been premature. Claims about necessity are often overblown because they offer highly restrictive and sometimes contradictory accounts of democratic outcomes. It is by pushing the boundaries of what is compared in terms of the depth, breadth and variety of cases that these issues come to the fore. It is better to qualify that most of these conditions are, as I will show in the following chapters, better thought of as INUS conditions:[14] they are necessary parts of relatively consistent explanations of the successful outcomes that are valid within alternate specific contexts. The finding has important implications for practice because it means that

[14] Insufficient, but necessary part of an unnecessary but sufficient condition.

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participatory innovation should not be dissuaded because a certain single condition seems beyond reach, or that all resources should be focused on some conditions because without them there is no prospect for success. There are alternative strategies for citizens to achieve democratic improvements through PB processes, and I now turn to describing them.

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